

PHP PDO + MySQL E-Commerce Project Scope

Project Name

ShopEase - E-Commerce Management System

Project Objective

Develop a complete E-Commerce application using **Core PHP, PDO, MySQL, Session, Cookie, OOP, and MVC Structure.**

The system should allow:

- Customer Registration & Login
 - Product Management
 - Category Management
 - Shopping Cart
 - Order Management
 - Admin Dashboard
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Technologies Used

- PHP 8+
 - MySQL
 - PDO
 - HTML5
 - CSS3
 - Bootstrap 5
 - JavaScript
 - jQuery
 - Session
 - Cookie
 - Git
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User Roles

1. Admin

- Manage Categories
- Manage Products
- Manage Orders
- Manage Customers
- Generate Reports

2. Customer

- Register/Login
- Browse Products
- Add to Cart
- Place Orders
- View Order History

Database Design

users table

```
id
name
email
phone
password
role
status
created_at
updated_at
```

categories table

```
id
name
slug
description
status
created_at
updated_at
```

products table

```
id
category_id
name
slug
```

description
price
discount_price
stock
image
status
created_at
updated_at

product_images table

id
product_id
image
created_at

carts table

id
user_id
product_id
quantity
price
created_at

orders table

id
user_id
order_no
subtotal
tax
shipping_charge
discount
grand_total
payment_method
payment_status
order_status
shipping_address
created_at

order_items table

id
order_id
product_id
quantity
price
total

wishlist table

```
id
user_id
product_id
created_at
```

Module 1: Authentication System

Features

Customer Registration

- Name
- Email
- Mobile
- Password
- Confirm Password

Login

- Email
- Password
- Remember Me

Logout

Module 2: Category Management (Admin)

Features

- Add Category
 - Edit Category
 - Delete Category
 - Search Category
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Module 3: Product Management (Admin)

Features

- Add Product

- Edit Product
 - Delete Product
 - Product Gallery
 - Product Search
 - Product Pagination
 - Product Status
-

Module 4: Product Listing (Frontend)

Features

- Product Details
 - Search Product
 - Category Wise Product
 - Price Filter
 - Latest Products
 - Related Products
-

Module 5: Shopping Cart

Features

- Add to Cart
 - Update Quantity
 - Remove Item
 - Empty Cart
 - Cart Total Calculation
-

Module 6: Wishlist

Features

- Add Wishlist
 - Remove Wishlist
 - Move to Cart
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Module 7: Checkout System

Features

- Shipping Address
 - Billing Address
 - Place Order
-

Module 8: Order Management

Customer

- Order History
- Order Details
- Cancel Order

Admin

- View Orders
 - Change Order Status
-

Module 9: Review & Rating System (optional)

Features

- Add Review
 - Update Review
 - Delete Review
 - Product Rating
-

Module 10: Admin Dashboard

Dashboard Statistics

- Total Users
- Total Orders
- Total Products
- Total Revenue
- Pending Orders

Reports Module

Sales Report

- Daily
- Monthly
- Yearly

Product Report

- Low Stock Products
- Top Selling Products

Customer Report

- New Customers
- Active Customers

Required PDO Concepts

- ✓ PDO Connection
- ✓ Prepared Statements
- ✓ Transactions
- ✓ Fetch Mode
- ✓ Bind Param
- ✓ Bind Value
- ✓ Exception Handling
- ✓ Pagination
- ✓ CRUD Operations

Required OOP Concepts

- ✓ Class & Object
 - ✓ Constructor
 - ✓ Inheritance
 - ✓ Interface
 - ✓ Static Method
 - ✓ Traits
 - ✓ Dependency Injection
-

Security Features

- ✓ Password Hashing

```
password_hash()  
password_verify()
```

- ✓ CSRF Protection

- ✓ SQL Injection Prevention (PDO Prepared Statement)

- ✓ XSS Prevention

```
htmlspecialchars()
```

- ✓ Session Management

- ✓ Secure File Upload
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Production-Level Bonus Features

Invoice Generation (PDF)

Email Notification

Order Tracking

Coupon System

AJAX Search

Soft Delete

Activity Log

API for Mobile Application